

MOKOPANE, South Africa

WMO: 682710

Lat: 24.1967S

Lon: 29.0067E

Elev: 1097

StdP: 88.82

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	6.2	-8.5	2.1	15.1	-6.4	2.5	15.5	7.4	16.0	5.4	17.5	0.8	70	0.296

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	34.5	18.0	33.1	18.0	31.9	18.0	21.5	27.6	21.0	27.1	20.6	26.6	2.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	16.9	23.1	19.5	16.3	22.9	18.9	15.7	22.7	68.3	27.8	66.4	27.0	64.7	26.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.0	4.2	DB	2.6	37.1	2.0	1.2	1.2	37.9	0.0	38.6	-1.1	39.3	-2.5	40.2	(4)
(5)			WB	-0.5	22.2	1.5	0.7	-1.6	22.7	-2.4	23.1	-3.3	23.5	-4.4	24.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	24.4	24.7	23.7	20.8	17.8	15.2	14.7	18.3	21.8	23.1	23.4	23.7	(6)	
	DBStd	4.30	2.22	2.10	2.18	2.56	2.39	2.05	2.05	2.91	3.26	3.29	3.06	2.58	(7)	
	HDD10.0	1	0	0	0	0	0	0	0	1	0	0	0	0	(8)	
	HDD18.3	306	0	0	0	7	37	97	113	37	8	3	3	1	(9)	
	CDD10.0	3992	445	411	424	325	242	156	146	256	354	406	402	426	(10)	
	CDD18.3	1257	187	177	166	82	21	2	1	35	113	151	154	168	(11)	
	CDH23.3	9457	1359	1234	1035	462	194	44	34	334	951	1307	1251	1251	(12)	
(13)	CDH26.7	3325	488	449	335	119	28	2	1	70	359	532	476	465	(13)	
(14)	Wind	WSAvg	1.9	1.9	1.9	1.9	1.7	1.5	1.5	1.4	2.0	2.3	2.4	2.3	1.9	(14)
(15) Precipitation	PrecAvg	499	89	82	51	25	9	5	2	3	6	38	99	91	(15)	
	PrecMax	782	197	217	125	95	38	32	16	11	37	78	272	137	(16)	
	PrecMin	308	40	4	8	0	0	0	0	0	1	0	2	37	(17)	
	PrecStd	130	43	59	33	23	12	8	4	3	8	21	56	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.8	34.6	34.0	31.9	29.3	26.5	26.1	30.2	34.2	36.0	35.8	35.4	(19)	
		MCWB	18.7	19.2	18.4	17.8	14.9	12.9	11.9	14.5	15.5	17.3	17.8	19.4	(20)	
	2%	DB	32.7	32.7	31.8	29.6	27.2	24.5	24.1	28.6	32.3	33.6	33.4	33.1	(21)	
		MCWB	19.3	19.1	18.7	17.1	14.3	12.2	11.7	13.9	15.3	17.0	17.7	19.1	(22)	
	5%	DB	31.1	31.1	30.1	27.8	25.5	22.9	22.6	26.9	30.6	31.7	31.4	31.2	(23)	
		MCWB	19.4	19.2	18.6	16.7	14.1	11.9	11.5	13.0	14.9	16.7	17.9	19.4	(24)	
	10%	DB	29.5	29.5	28.5	26.1	23.8	21.3	20.8	25.1	28.7	29.8	29.5	29.3	(25)	
		MCWB	19.4	19.2	18.5	16.4	13.8	11.6	11.1	12.7	14.5	16.5	17.8	19.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.1	21.5	20.0	17.2	14.6	14.0	16.2	17.7	19.7	21.1	21.7	(27)	
		MCDB	27.9	27.8	27.6	26.1	24.4	22.3	20.8	25.4	29.4	30.5	28.1	29.6	(28)	
	2%	WB	21.5	21.4	20.8	19.0	16.2	13.7	13.2	15.2	16.9	18.8	20.3	21.0	(29)	
		MCDB	27.1	27.1	26.7	25.5	23.0	21.1	20.6	24.8	27.4	28.3	27.4	28.2	(30)	
	5%	WB	20.9	20.8	20.2	18.3	15.4	13.0	12.4	14.4	16.3	18.2	19.6	20.5	(31)	
		MCDB	26.5	26.6	26.0	24.8	22.5	20.2	20.0	23.7	26.5	27.2	26.7	27.3	(32)	
	10%	WB	20.4	20.3	19.7	17.6	14.6	12.3	11.7	13.7	15.7	17.6	19.0	20.0	(33)	
		MCDB	25.9	26.1	25.3	23.8	21.8	19.4	19.2	22.5	25.5	26.4	25.9	26.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.0	10.2	11.0	13.9	14.5	14.0	13.9	13.2	11.9	10.9	9.9	(35)	
		MCDBR	12.9	12.6	12.7	13.6	15.8	16.5	16.0	15.9	15.3	14.6	13.7	13.2	(36)	
	5% WB	MCWBR	3.8	3.4	3.6	4.1	5.8	6.9	6.8	6.1	5.3	4.6	4.0	3.7	(37)	
		MCWBR	9.5	10.0	9.7	11.2	12.8	13.8	13.2	13.3	13.0	12.0	11.0	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.386	0.385	0.376	0.348	0.310	0.294	0.290	0.347	0.401	0.425	0.390	0.390	(40)	
		taud	2.376	2.377	2.388	2.429	2.490	2.513	2.509	2.334	2.214	2.189	2.334	2.359	(41)	
	Ebn at Noon	Ebn at Noon	958	944	925	907	905	902	919	893	885	896	947	955	(42)	
		Edh at Noon	131	128	121	106	91	85	88	115	141	153	136	133	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.61	6.30	5.65	4.86	4.54	4.04	4.36	5.12	5.94	6.37	6.53	6.52	(44)	
	RadStd	RadStd	0.52	0.58	0.50	0.41	0.19	0.24	0.11	0.24	0.32	0.42	0.56	0.44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page